

Urinary Tract Obstruction

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DEFINITIONS

Obstructive uropathy refers to the structural or functional changes in the urinary tract that impede normal urine flow. *Obstructive nephropathy* refers to the kidney disease caused by impaired flow of urine or tubular fluid. *Hydronephrosis* refers to dilation of the urinary tract. *Hydronephrosis* is not synonymous with *obstructive uropathy* because the former can occur without functional obstruction to the urinary tract and can be absent in established obstruction. Obstructive uropathy and nephropathy frequently coexist, and their management requires close collaboration between nephrologists and urologists. Some surgical aspects of obstruction to the urinary tract are discussed in [Chapter 63](#).

Obstructive uropathy is classified according to the site, degree, and duration of the obstruction. Acute or chronic obstruction can occur anywhere in the urinary tract and includes intrarenal causes (casts, crystals) and extrarenal causes. Acute or chronic obstruction is further subdivided into obstruction of the upper urinary tract (the urinary tract above the vesicoureteral junction, often unilateral) and the lower urinary tract (usually bilateral obstruction located below the vesicoureteral junction). Complete obstruction of the urinary tract is termed *high grade*, whereas partial or incomplete obstruction is termed *low grade*.

Unilateral obstruction in a patient with two normal kidneys will not result in significant reduction in glomerular filtration rate (GFR) because the contralateral kidney compensates. However, bilateral obstruction or the obstruction of a single functioning kidney will result in kidney failure. In acute urinary tract obstruction, changes are mainly functional, whereas structural damage to the kidney results from more chronic obstruction. The kidney with acute functional changes may recover after effective release of the obstruction, but structural changes may be permanent and lead to chronic kidney disease (CKD). Urinary tract obstruction is a major cause of acute kidney disease and CKD worldwide in children and adults.

ETIOLOGY AND PATHOGENESIS

The causes of obstructive uropathy affecting the upper and lower urinary tracts are summarized in [Boxes 61.1 and 61.2](#).

Congenital Urinary Tract Obstruction

Congenital urinary tract obstruction occurs most frequently in males, most commonly because of posterior urethral valves (PUV) or pelvi-ureteral junction (PUJ) obstruction. If obstruction occurs early during development, the kidney fails to develop and becomes dysplastic. If the obstruction is bilateral, there is a high mortality rate from kidney failure. If the obstruction occurs later in gestation and is low grade or unilateral, hydronephrosis and nephron loss will still occur, but kidney

function may be sufficient to allow survival, and such patients may not present until later in childhood. Mild PUJ obstruction may not manifest until adulthood and in some patients may be an incidental finding ([Fig. 61.1](#)). However, with increased use and improved sensitivity of antenatal scanning, congenital abnormalities of the urinary tract are now frequently identified early, allowing prompt postnatal (and in some cases antenatal) intervention to relieve the obstruction and hence preserve kidney function. Congenital causes of obstruction are discussed further in [Chapter 52](#).

Acquired Urinary Tract Obstruction

Acquired urinary tract obstruction may affect either the upper or lower urinary tract and can result from intrinsic or extrinsic causes. Intrinsic causes of obstruction may be intraluminal or intramural.

Intrinsic Obstruction

Intraluminal obstruction. Intrarenal intraluminal obstruction may result from tubular obstruction, such as the deposition of uric acid crystals after treatment of hematologic malignancies (tumor lysis syndrome), the precipitation of Bence Jones protein in myeloma, or the precipitation or crystal formation of certain drugs (e.g., sulfonamides, acyclovir, methotrexate, indinavir). Uncommonly, patients with glomerulonephritis such as immunoglobulin A (IgA) nephropathy may develop severe glomerular hematuria with tubular obstruction from erythrocytes and acute kidney injury (AKI) that typically resolves.

Extrarenal intraluminal obstruction in young adults is most commonly caused by urinary calculi (see [Chapter 60](#)). Calcium oxalate stones are the most common and typically cause intermittent acute unilateral urinary tract obstruction. Any stone formation is a significant and independent risk factor for CKD, although end-stage kidney disease (ESKD) because of urinary stones is rare.¹ Less common causes of urinary lithiasis, such as struvite stones, uric acid stones, and cystinuria are often bilateral and recurrent and therefore more likely to cause CKD. Kidney calculi lodge more commonly in the calyx, PUJ, or vesicoureteral junction and at the level of the pelvic brim. Surgical management of stones is discussed in [Chapters 60 and 63](#). Intraluminal obstruction can also result from a sloughed papilla after papillary necrosis or blood clots after macroscopic hematuria (“clot colic”). Papillary necrosis may occur in diabetes mellitus, sickle cell trait or disease, analgesic nephropathy, amyloidosis, and acute pyelonephritis. Clot colic can occur with bleeding from kidney tumors (benign and malignant) or arteriovenous malformations, after kidney trauma or surgery, and in patients with polycystic kidney disease.

Intramural obstruction. Intramural obstruction can result from either functional or anatomic changes. Functional disorders include adynamic ureteral segments (usually at the junction of the ureter with the pelvis or bladder) and neurologic disorders. The latter may result in a contracted (hypertonic) bladder or a flaccid (atonic) bladder,

BOX 61.1 Causes of Lower Urinary Tract Obstruction^a**Urethral Anatomic Causes**

- Urethral strictures: trauma, *postinstrumentation*, infections such as gonococcal urethritis, nongonococcal urethritis, tuberculosis
- Posterior urethral valves
- Stones
- Blood clots
- Periurethral abscess
- Phimosi
- Paraphimosis
- Meatal stenosis

Urethral Functional Causes

- Anticholinergic drugs, antidepressants, levodopa

Prostate

- *Benign prostatic hypertrophy*
- *Prostatic carcinoma*

Bladder Anatomic Causes

- *Bladder cancer*
- *Schistosomiasis* (*Schistosoma haematobium* infection)
- Bladder calculi
- Bladder trauma, pelvic fracture

Bladder Functional Causes

- *Neurogenic bladder*: spinal cord defects or trauma, diabetes, multiple sclerosis, Parkinson disease, cerebrovascular accidents

^aThe most common causes are in italics.

depending on whether the lesion affects upper or lower motor neurons, and may lead to impaired bladder emptying with vesicoureteral reflux. Bladder dysfunction is very common in patients with multiple sclerosis and after spinal cord injury and is also seen in diabetes mellitus, Parkinson disease, and after cerebrovascular accidents. Some drugs (anticholinergics, levodopa) can alter neuromuscular activity of the bladder and result in functional obstruction, especially if there is preexisting bladder outflow obstruction (e.g., prostatic enlargement).

Anatomic causes of intramural obstruction of the upper urinary tract include transitional cell carcinoma of the renal pelvis and ureter and ureteral strictures secondary to radiotherapy, previous stones or endoscopic stone surgery, or retroperitoneal surgery. Rarely, obstruction may result from ureteral valve malfunction, polyps, or strictures after therapy for tuberculosis. Intramural obstruction of the lower urinary tract can result from urethral strictures, which are usually secondary to instrumentation or previous urethritis, or malignant and benign tumors of the bladder. Infection with *Schistosoma haematobium* when the ova lodge in the distal ureter and bladder is a common cause of obstructive uropathy worldwide; up to 50% of chronically infected patients develop ureteral strictures and fibrosis, with calcifications and contraction of the bladder.

Extrinsic Obstruction

The most common cause of extrinsic compression in females is pressure from a gravid uterus on the pelvic rim; the right ureter is more commonly affected. It is usually asymptomatic, and the changes resolve rapidly after delivery. Rarely, bilateral obstruction and AKI may occur. Ureteral dilation frequently may be seen in pregnancy as a result of hormonal effects (especially progesterone) on smooth muscle, but this

BOX 61.2 Causes of Upper Urinary Tract Obstruction^a**Intrinsic Causes****Intraluminal**

- Intratubular deposition of crystals (uric acid, drugs)
- *Stones*
- Papillary tissue
- Blood clots
- Fungal ball

Intramural

- Functional: pelviureteral or vesicoureteral junction dysfunction
- Anatomic: tumors (benign or malignant)
- Infections, granulomas, strictures

Extrinsic Causes**Reproductive System**

- Cervix: *carcinoma*
- Uterus: *pregnancy*, *tumors*, prolapse, endometriosis, pelvic inflammatory disease
- Ovary: tumor, cysts
- Prostate: *carcinoma*

Vascular System

- Aneurysms: aorta, iliac vessels
- Aberrant arteries: pelviureteral junction
- Venous: ovarian veins, retrocaval ureter

Gastrointestinal Tract

- Crohn disease
- Pancreatitis
- Appendicitis
- Diverticulitis
- Tumors

Retroperitoneal Space

- Lymph nodes
- Fibrosis: idiopathic, drugs, inflammatory or immunoglobulin G4-related disease
- Tumors: primary or metastatic
- Hematomas
- Radiation therapy

Other

- *Surgical disruption or ureteral ligation*

^aThe most common causes are in italics.

does not indicate functional obstruction (see [Chapter 44](#)). Carcinoma of the cervix also may cause extrinsic obstruction secondary to direct extension of the tumor to involve the urinary tract. Other pelvic pathologic processes that can cause ureteral compression include benign and malignant uterine and ovarian masses, endometriosis, and pelvic inflammatory disease. Compression of the ureters outside the bladder also may occur with uterine prolapse. Rarely, inadvertent ureteral ligation may occur during surgical procedures, particularly those related to obstetrics and gynecology and colorectal surgery. Unilateral ligation may go undetected, but AKI will result from bilateral ligation.

In males, the most common cause of extrinsic obstruction of the lower urinary tract is benign prostatic hypertrophy. Carcinoma of the prostate can result in obstruction either from direct tumor extension to the bladder outlet or ureters or from metastatic spread.



Fig. 61.1 Intravenous Urogram Demonstrating Pelviureteral Junction Obstruction. The study was performed in a previously asymptomatic adult to investigate nonspecific right loin pain. There is unilateral dilation of the pelvicalyceal system. The ureter has not been visualized.

Retroperitoneal pathology may result in extrinsic obstruction of the ureters, as can metastases or extension of tumors from the cervix, prostate, bladder, colon, ovary, and uterus. Primary tumors of the retroperitoneum, such as lymphomas and sarcomas, commonly cause obstruction. Obstruction can also result from inflammatory conditions affecting the retroperitoneum, such as Crohn disease and large bowel diverticulitis. In Crohn disease, the obstruction is usually right-sided as a result of ileocecal disease. Less common pathologic processes include retroperitoneal fibrosis, in which thick fibrous tissue encases the ureters and draws them medially (Fig. 61.2). Retroperitoneal fibrosis may be idiopathic but can result from inflammatory aortic aneurysms, certain drugs (e.g., β -blockers, bromocriptine, methysergide), previous irradiation, trauma or surgery, and granulomatous disease (e.g., tuberculosis, sarcoidosis). Retroperitoneal fibrosis is also associated with IgG4-related disease (see Chapter 65), which typically presents with autoimmune pancreatitis, and elevated serum IgG4 levels suggest this diagnosis.² IgG4-related disease may be diagnosed after retroperitoneal biopsy material with an IgG4-positive plasma cell infiltrate, fibrosis with a whorled (cartwheel) appearance, and an obliterative venous phlebitis evident.³ The main differential diagnosis for idiopathic or benign retroperitoneal fibrosis is lymphoma or other malignant mimics and therefore consideration should be given to the need for biopsy. Computed tomography (CT) positron emission tomography (PET) can help with the distinction between benign retroperitoneal fibrosis and malignant disease.⁴

Ureteral compression may be a result of vascular abnormalities, including aneurysmal dilation of the aorta or iliac vessels, aberrant



Fig. 61.2 Retrograde Pyelogram Showing Idiopathic Retroperitoneal Fibrosis. Dilation of the pelvicalyceal system is clearly demonstrated. The ureters, however, are not dilated, and the left ureter can be seen displaced medially as a result of being encased in thick fibrous tissue.

vessels such as an aberrant anterior crossing accessory renal artery causing PUJ obstruction in adults, and anatomic variations in the location of the ureter (retrocaval ureter).

Pathophysiology

Obstruction of the urinary tract causes profound functional and structural changes of the kidney. Initially, changes are predominantly functional and potentially reversible, but with time, chronic and irreversible structural changes occur. Our understanding of the consequences of urinary tract obstruction stems mainly from the study of animal models.⁵ Although many studies have focused on the effects of complete ureteral obstruction in rodents, investigators have also examined models of chronic complete, partial, or reversible obstruction in adult and neonatal animals.⁵ Available experimental data show little species-to-species variation in the response to acute obstruction, suggesting that similar changes are likely to occur in humans. The complex effects of urinary tract obstruction on the kidney affect both glomerular hemodynamics and tubular function.

Changes in Glomerular Function

GFR declines progressively after the onset of complete ureteral obstruction. After complete ureteral obstruction, there is an initial rise in proximal tubular pressure. This is accompanied by afferent arteriolar dilation as a result of the generation of vasodilatory prostaglandins. Although this increases glomerular capillary hydraulic pressure, it does not offset the rise in tubular pressure, and there is a decrease in the hydraulic pressure gradient across glomerular capillaries, resulting in a dramatic decline in GFR. The relative changes in ureteral pressure, renal plasma flow, and GFR are summarized in Fig. 61.3. With ongoing obstruction, there is a progressive fall in renal blood flow secondary to the generation of angiotensin II (Ang II) and thromboxane A_2 , the release of vasopressin (antidiuretic hormone), and decreased nitric oxide production. An interstitial leukocyte infiltrate develops, predominantly macrophages, and promotes the late structural changes that occur after obstruction as macrophage depletion limits experimental interstitial fibrosis.⁶

The extent to which glomerular function recovers after the release of ureteral obstruction depends on the duration of the obstruction. Whole-kidney GFR may return to normal after short-term obstruction (days), whereas recovery may be incomplete after prolonged obstruction.

Changes in Tubular Function

Abnormalities in tubular function are common in urinary tract obstruction and manifest as altered kidney handling of electrolytes and changes in the regulation of water excretion. The degree and nature of the tubular defects after obstruction depend in part on whether the obstruction is bilateral or unilateral. These differences could result from the dissimilar hemodynamic responses, different intrinsic changes within the nephron, or differences in extrinsic factors (e.g., volume expansion and accumulation of natriuretic substances in bilateral obstruction) between the two states.

After ureteral obstruction, the ability to concentrate the urine is impaired with causative factors including a loss of medullary tonicity and reduced expression of sodium transporters. Also, the collecting duct is unresponsive to vasopressin because of reduced expression of kidney aquaporins.

Rats exhibit reduced expression of multiple acid-base transporters after ureteral obstruction,⁷ and patients with urinary tract obstruction often have urinary acidification defects. These defects may be detected only by exogenous acid loading, but hyperchloremic acidosis caused by impaired distal acid secretion, hyporeninemic hypoaldosteronism (type 4 renal tubular acidosis), and a combination of these findings has been described. This acidifying defect results from a marked increase in bicarbonate excretion or from a distal acidification defect, possibly as a result of abnormalities of the H⁺-ATPase activity of intercalated cells of the collecting duct after ureteral obstruction.

Obstruction also alters kidney potassium handling. In the presence of a normal functioning contralateral kidney, potassium excretion is reduced after relief of obstruction, either in proportion to or perhaps even greater than the fall in GFR (i.e., fractional excretion of potassium is unaltered or slightly reduced). There is a defect in the distal potassium secretory mechanism after unilateral obstruction that may be secondary to reduced responsiveness of that nephron segment to aldosterone. By contrast, after release of bilateral ureteral obstruction,

there is a marked increase in both net and fractional potassium excretion. The major mechanism by which potassium losses occur in this setting is an increased delivery of sodium to the distal tubule, resulting in increased sodium-potassium exchange.

Recovery of tubular function after release of obstruction is slow and may remain abnormal even after whole-kidney GFR has returned to normal. In rats, acidification and potassium-handling abnormalities persist for at least 14 days and urinary concentrating ability is abnormal for up to 60 days after the release of 24 hours of unilateral ureteral obstruction. These observations are consistent with persistent alterations in distal tubular and collecting duct function or a loss in functioning juxtaglomerular nephrons after the release of the obstruction.

HISTOPATHOLOGIC CHANGES

The morphologic alterations in kidney architecture are similar irrespective of the cause of the obstruction. Initially, there is kidney enlargement and edema with pelvicalyceal dilation (Fig. 61.4). Tubular dilation that predominantly affects the collecting duct and distal tubular segments is seen microscopically with cellular flattening and atrophy of proximal tubular cells. Glomerular structures are usually preserved initially, although the Bowman space may be dilated and some periglomerular fibrosis may ultimately develop.

Inadequately treated urinary tract obstruction eventually causes irreversible structural changes. The renal pelvis becomes widely dilated, with the papillae either flattened or hollowed out. The cortex and medulla become grossly thinned, such that the kidney becomes a thin rim of kidney tissue surrounding a large saccular pelvis (Fig. 61.5). Histologic examination demonstrates tubulointerstitial fibrosis and profound nephron loss. Tubular proliferation and apoptosis, interstitial myofibroblast accumulation, increased extracellular matrix deposition, and tubular atrophy occur. Ischemia as a result of the decreased renal blood flow and the rarefaction of peritubular capillaries contributes to parenchymal damage after obstruction. In both genetic and interventional studies, an important pathologic role for Ang II and transforming growth factor- β (TGF- β) has been established.⁵

Infiltrating macrophages release profibrogenic factors (e.g., TGF- β , galectin-3) and play a pivotal role in the chronic tissue injury and fibrosis that result from prolonged ureteral obstruction^{6,8} (Fig. 61.6). Local Ang II generation also may stimulate tubular cell production of TGF- β . Treatments shown to ameliorate chronic interstitial damage in experimental obstructive uropathy include angiotensin receptor

The Effects of Complete Ureteral Obstruction

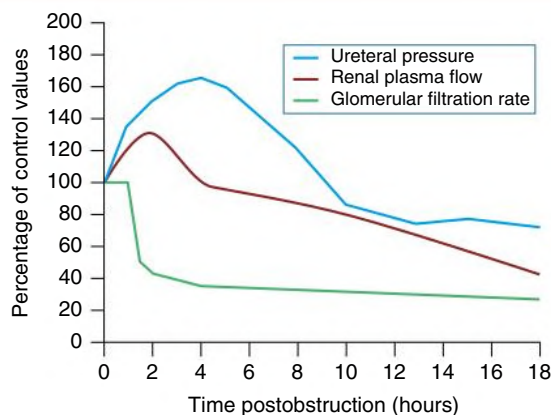


Fig. 61.3 Effects of Complete Ureteral Obstruction. The relative changes in ureteral pressure, kidney plasma flow, and glomerular filtration rate are shown using data from experimental studies of unilateral ureteral obstruction in rats.



Fig. 61.4 Autopsy Specimen of a Kidney Showing the Early Effects of Ureteral Obstruction. The kidney is enlarged and edematous with pelvicalyceal dilation. There is good preservation of the kidney parenchyma.



Fig. 61.5 Chronic Ureteral Obstruction. Surgical specimen of a kidney showing gross dilation of the pelvicalyceal system and the reduction of the kidney cortex to a thin fibrotic rim of tissue. There would have been no prospect for any significant functional recovery in this kidney after the relief of the obstruction.

blockers (ARBs), pentoxifylline, simvastatin, and various growth factors with beneficial effects including reduced tubulointerstitial inflammation, tubular cell apoptosis, and fibrosis.⁵ However, differences have been noted in the responses of adult and neonatal rodents to experimental therapeutic interventions, and it is unclear whether such differences might occur in humans.

EPIDEMIOLOGY

Obstructive uropathy is common and can occur at all ages, with the prevalence of hydronephrosis at autopsy being 3.5% to 3.8%, although this underestimates the true incidence because these figures exclude transient obstruction. The frequency and cause of obstruction vary in both sexes with age. Antenatal ultrasound has significantly increased the detection rate of lower urinary tract obstruction in the fetus.⁹ In children aged younger than 10 years, obstruction is more common in males; congenital urinary tract anomalies, such as PUV and pelviureteral obstruction, account for most cases. In North America, obstructive uropathy is a common cause of ESKD in children and accounts for 11.4% of cases.¹⁰ In addition, congenital obstructive uropathy accounts for 0.7% of all patients (median age, 31 years) treated with maintenance kidney replacement therapy,

Impairment of Function and Structural Damage in Obstructive Nephropathy

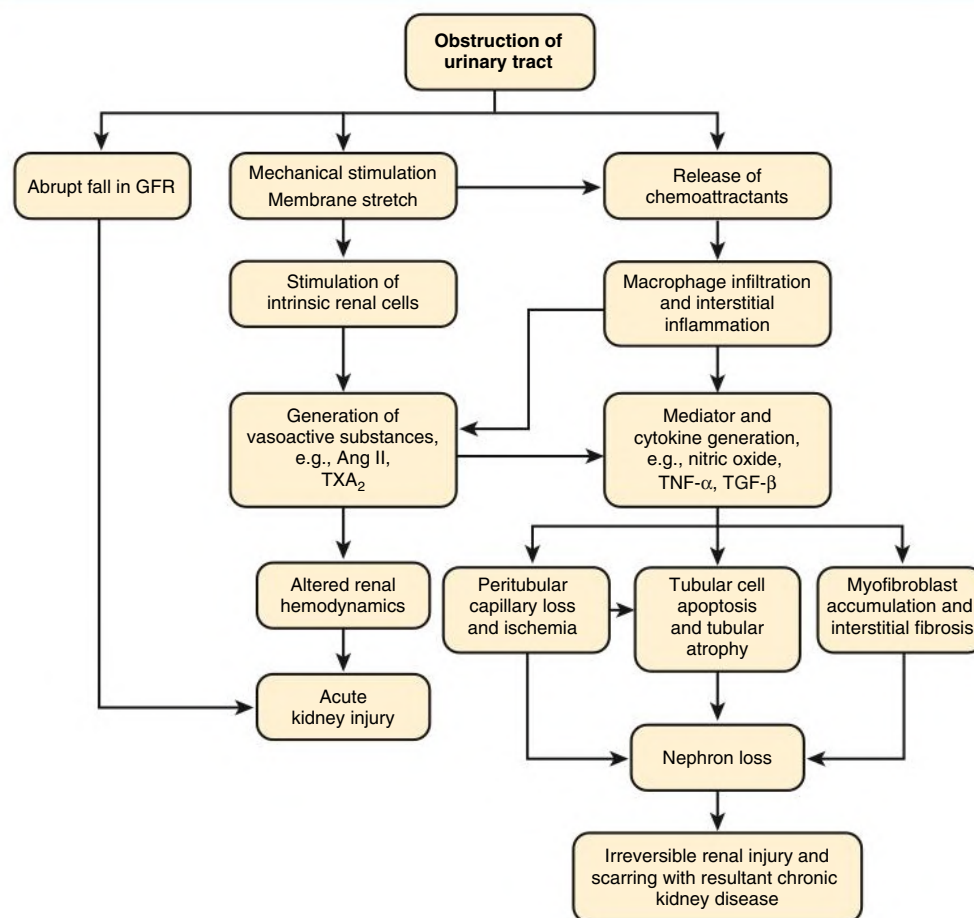


Fig. 61.6 Events leading to acute impairment of kidney function and chronic structural damage in obstructive nephropathy. *Ang II*, Angiotensin II; *GFR*, glomerular filtration rate; *TGF-β*, transforming growth factor-β; *TNF-α*, tumor necrosis factor α; *TXA₂*, thromboxane A₂.

demonstrating the continued impact of this disease into adult life.¹¹ In adults younger than 20 years, the frequency of urinary tract obstruction is similar in males and females. Beyond 20 years of age, obstruction becomes more common in females, mainly because of pregnancy and gynecologic malignancy. The peak incidence of kidney calculi occurs in the second and third decades of life, with a threefold increased incidence in males. After the age of 60 years, obstructive uropathy occurs more frequently in males secondary to benign prostatic hypertrophy and prostatic carcinoma. About 30% of males older than 50 years have some symptoms of bladder outflow obstruction and acquired urinary tract obstruction can lead to ESKD in elderly patients with most resulting from prostatic disease.¹² In the United States, 1.4% of prevalent patients with ESKD have kidney failure secondary to acquired obstruction.¹¹

CLINICAL MANIFESTATIONS

Obstruction of the urinary tract can present with a wide range of clinical symptoms, depending on the site, degree, and duration of obstruction. The clinical manifestations of upper and lower urinary tract obstruction differ. Symptoms can be caused by mechanical obstruction of the urinary tract (usually pain) or can result from the complex alterations in glomerular and tubular function that may occur in obstructive nephropathy. The latter commonly manifests as alterations in urine volume and as kidney failure, which can be acute or chronic. For example, patients with complete obstruction present with anuria and AKI, whereas those with partial obstruction may present with polyuria and polydipsia secondary to acquired vasopressin resistance. Alternatively, there may be a fluctuating urine output, alternating from oliguria to polyuria. However, obstructive uropathy and hence obstructive nephropathy can occur without symptoms and with minimal clinical manifestations. Therefore, obstruction of the urinary tract must be considered in the differential diagnosis of any patient with reduced GFR.

Pain

Pain is a frequent complaint in patients with obstructive uropathy, particularly in those with ureteral calculi. The pain is believed to result from stretching of the collecting system or the renal capsule. The location of the pain may help determine the site of obstruction. With upper ureteral or pelvic obstruction, flank pain and tenderness typically occur, whereas lower ureteral obstruction causes pain that radiates to the groin, the ipsilateral testicle, or the labia. Acute high-grade ureteral obstruction may be accompanied by a steady and severe crescendo flank pain radiating to the labia, the testicles, or the groin (classic kidney colic). The acute attack may last less than half an hour or as long as a day. By comparison, patients with chronic slowly progressive obstruction may have no pain or minimal pain; in such patients, any pain that does occur is rarely colicky. In PUJ obstruction, pain may occur only after fluid loading to promote a high urine flow rate (often reported after drinking alcohol, for example).

Lower Urinary Tract Symptoms

Obstructive lesions of the urethra or bladder neck or bladder disease may cause a decrease in the force or caliber of the urine stream, intermittency, hesitancy, straining (voiding lower urinary tract symptoms), postmicturition dribbling, or nocturia. Urgency, frequency, and urinary incontinence can result from incomplete bladder emptying. Indeed, the development of nocturnal incontinence suggests chronic urinary retention and should be considered a red flag that may be associated with painless kidney failure. Such symptoms commonly result from prostatic hyperplasia (a histologic diagnosis; benign prostatic

enlargement [BPE] is the preferred clinical term) and are frequently referred to as *prostatism*. The preferred clinical term is lower urinary tract symptoms (LUTS), caused by BPE, resulting in benign prostatic obstruction (BPE with evidence of bladder obstruction). However, lower urinary tract symptoms have many causes and are not pathognomonic of benign prostatic enlargement.

Urinary Tract Infections

Urinary stasis resulting from obstruction predisposes to urinary tract infections (UTIs), and patients may develop cystitis with dysuria and frequency or pyelonephritis with loin pain and systemic symptoms. Infection occurs more often in patients with lower urinary tract obstruction than in those with upper urinary tract obstruction.

UTI in males or young children of either sex, recurrent or persistent infections in females, infections with unusual organisms such as *Pseudomonas* or *Proteus spp.*, and a single attack of acute pyelonephritis require further investigation to exclude obstruction or stone disease. Also, the presence of obstruction makes effective eradication of the infection difficult. Infections of the urinary tract with a urease-producing organism such as *Proteus mirabilis* predispose to stone formation. These organisms generate ammonia, which results in urine alkalization and favors the development of magnesium ammonium phosphate (struvite) stones. Struvite calculi can fill the entire renal pelvis to form a staghorn calculus that eventually leads to loss of the kidney if untreated. Thus, stone formation and papillary necrosis can be consequences of urinary tract obstruction and causes of obstruction.

Hematuria

Calculi may cause trauma to the urinary tract uroepithelium and result in either macrohematuria or microhematuria. Any neoplastic lesion that obstructs the urinary tract, especially uroepithelial malignancies, may bleed, resulting in macrohematuria. Urinary tract bleeding may result in obstruction, giving rise to clot colic when it is in the ureter, or clot retention when it is in the bladder. Clot obstruction of the ureter is usually (but not always) transient.

Changes in Urine Output

Complete bilateral obstruction or unilateral obstruction of a single functioning kidney such as a kidney transplant will result in anuria. However, in partial obstruction, the urine output may be normal or increased (polyuria). A pattern of alternating oliguria and polyuria or the presence of anuria strongly suggests obstructive uropathy.

Abnormal Physical Findings

Physical examination can be completely normal. Some patients with upper urinary tract obstruction may have flank tenderness. Long-standing obstructive uropathy may result in an enlarged palpable kidney in children. Lower urinary tract obstruction causes a distended, palpable, and occasionally painful bladder. However, in severe chronic bladder outflow obstruction (which is most likely to cause kidney failure), there may be no tenderness and no discrete palpable bladder because the bladder has stretched to such an extent that it is now “baggy.” Ultrasound scan, or catheterization with careful documentation of residual bladder volume and subsequent urine output, may be required to clinch the diagnosis. A rectal examination and, in females, a pelvic examination should be performed because they may reveal a local malignancy, prostatic enlargement, or neurologic deficit suggesting cauda equina pathology.

Acute or chronic hydronephrosis, either unilateral or bilateral, may cause hypertension secondary to impaired sodium excretion with expansion of extracellular fluid volume or from the abnormal release

of renin. On occasion, in patients with partial urinary tract obstruction, polyuria and volume depletion lead to hypotension.

Abnormal Laboratory Findings

Urinalysis may show hematuria, bacteriuria, pyuria, crystalluria, and low-grade proteinuria, depending on the cause of obstruction. However, urinalysis may be completely negative despite advanced obstructive nephropathy. In the acute phase of obstruction, urinary electrolyte values are similar to those seen in a prerenal state, with a low urinary sodium concentration (<20 mmol/L), a low fractional excretion of sodium ($<1\%$), and a high urinary osmolality (>500 mOsm/kg). However, with more prolonged obstruction, a decreased ability to concentrate the urine and an inability to reabsorb sodium and other solutes occur. These changes are particularly marked after the release of chronic obstruction and give rise to the syndrome commonly referred to as “postobstructive diuresis.”

Increases in serum urea and creatinine are the most significant laboratory abnormalities in patients with obstructive uropathy. Electrolyte abnormalities may occur, including hyperchloremic hyperkalemic (type 4) metabolic acidosis or hyponatremia from acquired nephrogenic diabetes insipidus. The development of obstruction in patients with underlying CKD may accelerate the rate of progression. ESKD may occasionally be caused by chronic obstructive uropathy that had been asymptomatic.

Obstruction in Neonates or Infants

With the advent of routine antenatal scanning, the diagnosis of hydronephrosis and genitourinary abnormalities is now frequently made antenatally; however, unsuspected obstructive uropathy may present in the postnatal period with failure to thrive, voiding difficulties, fever, hematuria, or symptoms of kidney failure. Oligohydramnios at the time of delivery should raise the suspicion of obstructive uropathy, as should the presence of congenital anomalies of the external genitalia. Nonurologic anomalies such as ear deformities, a single umbilical artery, imperforate anus, or a rectourethral or rectovaginal fistula should prompt investigation for urinary tract obstruction. Any infant with neurologic abnormalities may have a neurogenic bladder with associated obstructive uropathy.

DIAGNOSIS

Prompt diagnosis of urinary tract obstruction is essential to allow timely treatment. Symptoms such as kidney colic may suggest the diagnosis and prompt appropriate investigation. However, urinary tract obstruction should be considered in any patient with unexplained acute or chronic kidney impairment. The diagnostic approach must be tailored to the clinical presentation (Fig. 61.7), but a careful history and thorough physical examination are mandatory in all patients.

Urinalysis may provide valuable diagnostic information. Hematuria suggests that the obstructing lesion is a calculus, sloughed papilla, or tumor. Bacteriuria suggests urinary stasis, especially in males or pregnant females, but it also may be a complication of chronic obstruction. The presence of crystals in the urine sediment (cystine or uric acid) may be an indication of the type of stone causing the ureteral obstruction or the intrarenal obstruction resulting in AKI. Laboratory studies must include assessment of GFR and serum electrolytes.

Imaging

Because the sites, causes, and consequences of obstruction to the urinary tract are so variable, no single imaging investigation can diagnose or exclude urinary tract obstruction with certainty. Therefore, if the clinical suspicion of obstruction is high, the patient may require investigation with multiple imaging techniques.

Ultrasound is the most widely used imaging modality, but CT scanning (with urographic phase) and magnetic resonance (MR) urography are increasingly used to accurately diagnose both the site and cause of obstruction. Although much less commonly used, older imaging techniques, such as intravenous urography (IVU), can be used to evaluate patients with obstructive uropathy. None of these modalities provides a functional kidney excretory or voiding profile, and where doubt still exists as to the existence of kidney obstruction (e.g., in cases of nonobstructive hydronephrosis [“baggy renal pelvis”], vesicoureteral reflux, or hypertonic bladder), functional investigations are required such as radioisotope excretory renography (mercaptoacetyltriglycine [MAG3] renogram), micturating cystoureterogram, or pressure-voiding studies (cystometrogram [CMG]), respectively. The role of imaging techniques is shown in Fig. 61.7 and discussed further in Chapters 5 and 6.

Ultrasound

Ultrasound can define kidney size and demonstrate calyceal dilation but depends on the expertise of the operator (Fig. 61.8). Although it is sensitive for detection of hydronephrosis, ultrasound often will not determine its cause. Pathologic change within the ureter is difficult to demonstrate, and tiny stones will not generate acoustic shadows. However, unilateral hydronephrosis suggests obstruction of the upper urinary tract by stones, blood clots, or tumors. Bilateral hydronephrosis is more likely to result from a pelvic problem obstructing both ureters or obstruction of the bladder outlet, in which case the bladder will also be enlarged. Ultrasound should be combined with radiographic examination of the kidneys, ureters, and bladder (KUB) to ensure that ureteral stones or small kidney stones are not overlooked.

Ultrasound produces false-negative results in cases of nondilated obstructive uropathy. Immediately after acute obstruction (within 24 hours), the relatively noncompliant collecting system may not have dilated, so ultrasound examination findings may be normal. Furthermore, if urine flow is low, as in severe dehydration or kidney failure, there may be little dilation of the urinary tract. Dilation also may be absent in slowly progressive obstruction when the ureters are encased by fibrous tissue (as in retroperitoneal fibrosis) or by tumor, and in some kidney transplants in which there is dense surrounding scar tissue. The acoustic shadow of a staghorn calculus also can mask dilation of the upper urinary tract. In view of this, it is advisable to repeat the kidney ultrasound if suspicion of obstruction is high. The sensitivity of ultrasound for diagnosis of obstruction can be improved by measuring the resistive index with color Doppler sonography. A resistive index above 0.7 reflects the increased vascular resistance present in obstruction and can assist discrimination between obstructed and nonobstructed kidneys, although the resistive index can be increased in other conditions.¹³ Such ultrasound techniques are particularly useful when it is especially important to minimize radiation exposure, such as in pregnant patients and in children, and in the follow-up of patients requiring repeated imaging, such as after extracorporeal shock wave lithotripsy.

Even in experienced hands, ultrasound may have a significant false-positive rate, especially if minimal criteria are adopted to diagnose obstruction. In addition, the echogenicity produced by multiple kidney cysts may be mistaken for hydronephrosis.

Ultrasound scanning can be used to assess bladder emptying and should be undertaken in patients with lower urinary tract symptoms, being performed at the bedside if necessary. The normal postmicturition (postvoid) residual volume is less than 50 mL (<100 mL is usually acceptable in patients >65 years), and a large postmicturition residual volume suggests bladder outflow obstruction or detrusor muscle failure, which should prompt further urologic investigation and treatment.

Investigation and Management of Suspected Urinary Tract Obstruction

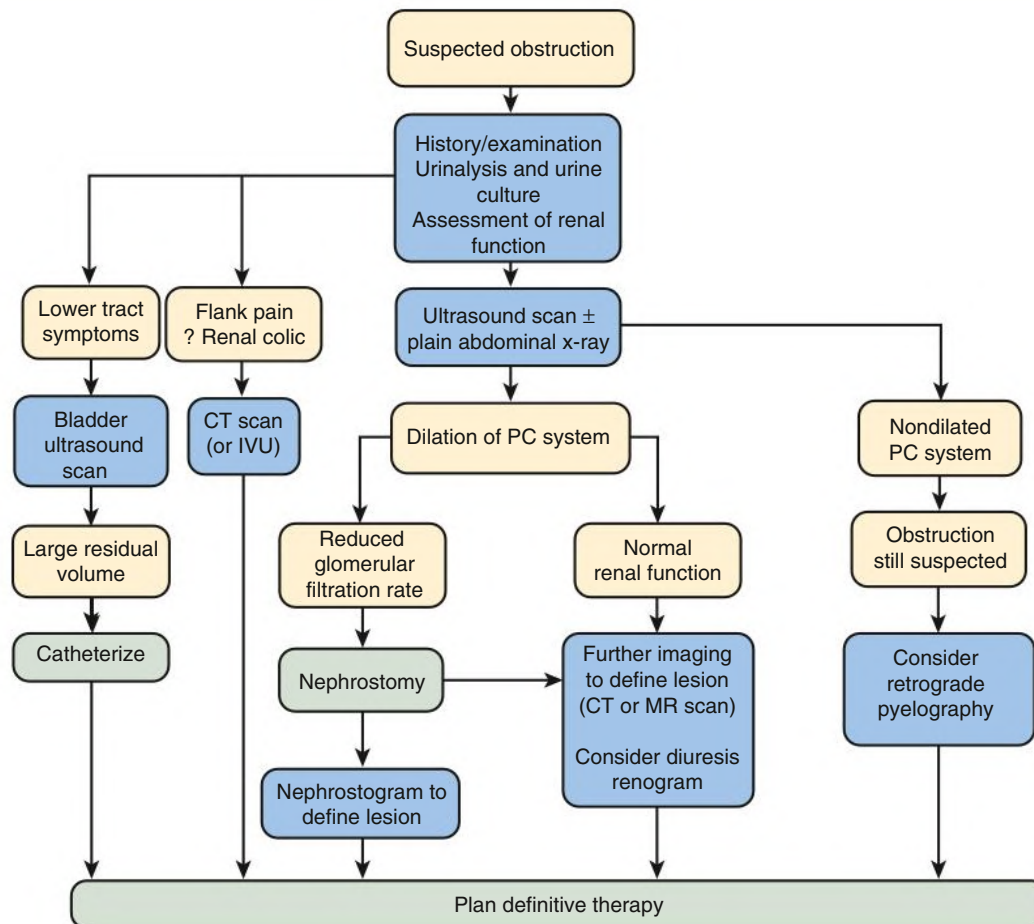


Fig. 61.7 Investigation and Management of Suspected Urinary Tract Obstruction. A full history and examination should be performed, together with urinalysis, urine microscopy and culture, and measurement of kidney function and serum electrolytes. Ultrasound is a useful first-line investigation for any patient with suspected urinary tract obstruction. Computed tomography (CT) is now the preferred imaging technique when kidney calculi are suspected. Either CT or magnetic resonance (MR) urography can accurately diagnose both the site and cause of obstruction in most cases. If there is reduced glomerular filtration rate, a nephrostomy or stenting allows the effective relief of the obstruction and time for kidney function to recover while definitive therapy is planned. IVU, Intravenous urography; PC, pelvicalyceal.

The investigation of neonates with hydronephrosis diagnosed antenatally depends on the grade of hydronephrosis identified. Neonates with grade 1 or 2 hydronephrosis (no calyceal dilation) undergo ultrasound scanning; neonates with grade 3 or 4 hydronephrosis (indicating increasingly severe pelvicalyceal dilation) require both ultrasound scanning and voiding cystourethrography.¹⁴ This combination can distinguish megaureter resulting from obstruction or reflux and enable the diagnosis of PUV and UPJ obstruction. Measurements of both calyceal dilation and the anteroposterior diameter of the kidney pelvis at the first postnatal ultrasound may be able to distinguish those children who will require surgery, but this grading system will require further validation.¹⁵

Plain Abdominal Radiography

A plain abdominal radiograph (or KUB) allows an assessment of kidney size and contour and frequently demonstrates kidney calculi because about 90% of calculi are radiopaque.

Intravenous Urography

IVU was formerly the first-choice investigation for suspected upper urinary tract obstruction. In patients with normal GFR, it can usually define both the site and the cause of the obstruction. However, the excretion of contrast material may be poor or delayed in patients with low GFR because of a decreased filtered load of the contrast agent. IVU is no longer a first-line investigation to diagnose urinary tract obstruction, especially in patients with reduced GFR.

Computed Tomography

Non-contrast-enhanced spiral CT scanning is used increasingly as the primary imaging modality for the evaluation of patients with acute flank pain¹⁶ and uses a low-dose protocol compared with conventional CT scanning of abdomen and pelvis. Stones are easily detected because of their high density; CT can provide an accurate and rapid diagnosis of an obstructing ureteral calculus, with a stone being found more commonly in males.¹⁷ In addition, it provides useful information

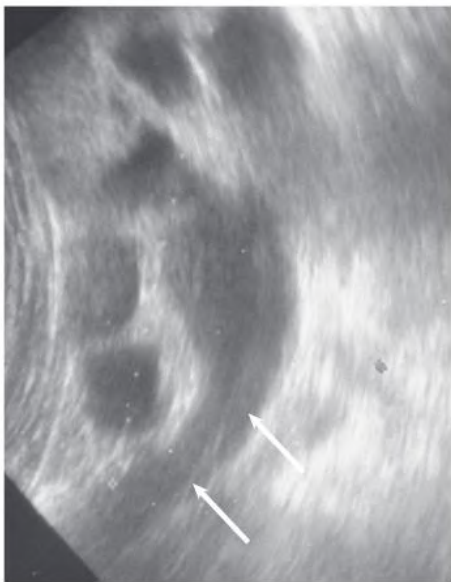


Fig. 61.8 Kidney Ultrasound Scan of a Patient with Obstruction of the Urinary Tract Causing Hydronephrosis. The kidney is hydronephrotic with dilation of the pelvicalyceal system; dilation of the upper ureter is also clearly seen (arrows).

about the site and nature of the obstructing lesion, especially when this is extrinsic to the urinary tract (Fig. 61.9), although a higher dose CT scan with intravenous (IV) contrast may be required for a full diagnostic study. CT demonstrates retroperitoneal disease, such as para-aortic and paracaval lymphadenopathy; retroperitoneal fibrosis is evident as increased attenuation within the retroperitoneal fat, with encasement of one or both ureters. Hematomas, primary ureteral tumors, and polyps are also detectable. The diagnostic potential of CT is enhanced by the administration of contrast material, but this may limit its use in patients with reduced GFR. CT urogram (IV contrast with delayed [usually 10 minutes] excretory phase images) is the gold standard imaging modality for diagnosis of intraluminal or intramural obstructing lesions such as upper tract urothelial malignancy. It should be noted that CT involves exposure to ionizing radiation, especially if patients require multiple CT scans over time.

Magnetic Resonance Urography

MR urography (combined with KUB) can enable the diagnosis of ureteral obstruction caused by kidney calculi with accuracy similar to that of spiral CT scanning but without exposure to contrast medium or ionizing radiation. The technique has less observer variability and is more accurate than CT in detecting indirect evidence of obstruction, such as perirenal fluid. MR urography can rapidly and accurately depict the morphologic features of dilated urinary tracts and provide information about the degree and level of obstruction¹⁸ (Fig. 61.10). MR urography is a particularly attractive imaging modality for the evaluation of hydronephrosis in children because it provides both anatomic and functional data and can indicate whether the hydronephrosis is compensated (symmetric changes of signal intensity of the nephrogram) or decompensated.¹⁹ Signs of decompensation (acute on chronic obstruction) include edema of the kidney parenchyma, a delayed and increasingly dense nephrogram, a delayed calyceal transit time, and a more than 4% difference in the calculated differential kidney function. MR urography is likely to be increasingly used in the future.

Retrograde Pyelography

Retrograde pyelography (Fig. 61.11; see also Fig. 61.2) usually requires a general anesthetic and cystoscopy but may be particularly useful to



Fig. 61.9 Computed Tomography Scan of the Abdomen Showing a Grossly Hydronephrotic Kidney on the Left. Arrows mark dilated kidney pelvis. Dilated loops of small bowel are seen in the right hypochondrium. Sequential sections demonstrated that the ureter was dilated along its length and that there was a pelvic mass, which was responsible for both bowel and left ureteral obstruction. The mass was subsequently shown to be arising from a carcinoma of the colon.

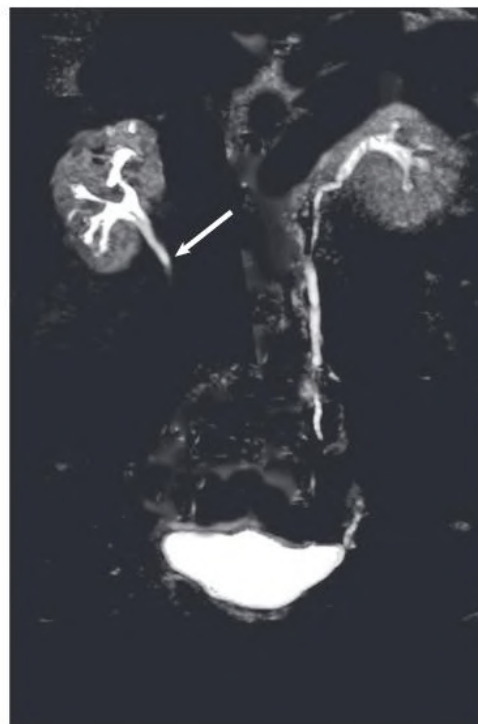


Fig. 61.10 Magnetic Resonance (MR) Urography Showing Obstructive Uropathy. T2-weighted MR image showing a proximal right-sided ureteric ureteral obstruction with an associated mild hydronephrosis. The obstruction was secondary to a ureteral calculus.

identify both the site and the cause of the obstruction because intervention (by stenting or dilation) and biopsy of a suspected tumor (by ureteroscopy) are both possible. It is also helpful when nondilated urinary tract obstruction is suspected or when there is a history of allergic reactions to contrast material.

Diuresis Renography

A diuresis renogram using technetium-99m MAG3 (^{99m}Tc-MAG3), combined with IV furosemide administered 20 to 30 minutes after injection of the isotope (diuretic isotopic renography), can be used to distinguish between simple dilation of the collecting system and true



Fig. 61.11 Ureteral Obstruction by a Tumor. A retrograde pyelogram shows the tumor is within and obstructing the ureter (*arrows*). Above the tumor, there is dilation of the ureter, but below it, the ureter is of a normal caliber.

obstruction.²⁰ Normally, there is a rapid washout of the isotope from the kidney, and persistence of the isotope suggests a degree of obstruction (Fig. 61.12). Severely reduced GFR significantly limits the usefulness of renography because the diuretic response to furosemide may be absent. Diuresis renography also may be used for follow-up of patients who have undergone surgical procedures to relieve obstruction, such as a pyeloplasty.

Pressure Flow Studies

A pressure flow study (Whitaker test) involves puncture of the collecting system with a fine-gauge needle to perfuse fluid (at 10 mL/min) with concurrent measurement of the differential pressure between the bladder and the collecting system; a pressure greater than 20 cm H₂O indicates obstruction. Although this test is rarely required, it may be informative in patients with potential upper tract obstruction when other, less invasive tests have generated equivocal results.²¹

Other Evaluations

Lower urinary tract obstruction may be evaluated by cystoscopy, which allows a visual inspection of the entire urethra and the bladder. Urodynamic studies, also known as cystometrograms (CMG; see Chapter 50, Fig. 50.14), are most useful where the distinction between obstructive (e.g., BPE) and functional (e.g., detrusor muscle failure) impaired bladder emptying needs to be determined. High pressure voiding and vesicoureteral reflux can also be assessed with the use of fluoroscopy during urodynamics (known as video urodynamics).

Diuretic Isotopic Renography

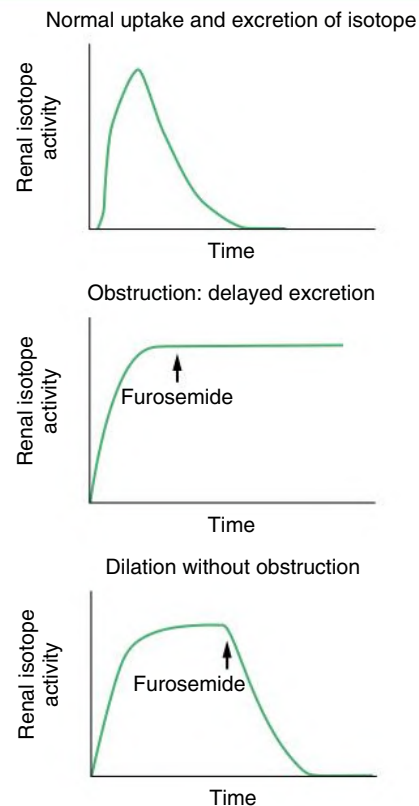


Fig. 61.12 Diuretic Isotopic Renography. Idealized tracings for normal, obstructed, and dilated kidneys without obstruction of the upper urinary tract. In obstruction, there is delayed excretion of technetium-99m mercaptoacetyltriglycine despite administration of furosemide. When there is dilation of the upper urinary tract without obstruction, the isotope is retained but is rapidly excreted after the administration of furosemide.

DIFFERENTIAL DIAGNOSIS

Diagnostic uncertainty arises with nonobstructive dilation of the upper urinary tract, which may be seen with vesicoureteral reflux, diuretic administration, diabetes insipidus, congenital megacalyces, chronic pyelonephritis, and postobstructive atrophy. Diuresis renography or retrograde pyelography may be required to exclude obstruction.

NATURAL HISTORY

Obstructive uropathy is potentially curable but will result in progressive irreversible loss of nephrons and kidney scarring if left untreated (Fig. 61.13). ESKD will result if both kidneys are affected or if there is only a solitary kidney. Outcome data for obstructive uropathy are limited, but the exact prognosis will depend on the pathologic process responsible for the obstruction, the duration and grade of the obstruction (complete or partial), and the presence or absence of urosepsis. Relief of short-term obstruction (<1–2 weeks) usually results in an adequate return of kidney function. With chronic progressive obstruction (>12 weeks), there is often irreversible and severe kidney damage, and kidney functional recovery may be limited even after relief of the obstruction. A single-center study identified 104 patients who presented with obstructive nephropathy.²² The mean GFR at presentation and at 3, 12, and 36 months was 9, 28, 29, and 30 mL/min, respectively

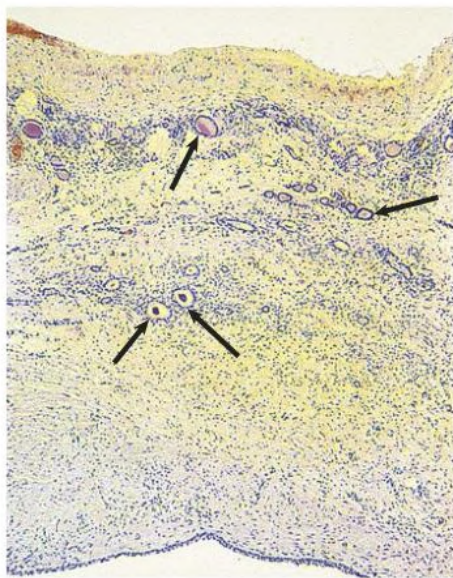


Fig. 61.13 Pathology of Chronic Ureteral Obstruction. Section of the rim of kidney tissue from the kidney shown in Fig. 61.7. The kidney capsule is at the top, the urinary space at the bottom. The cortex is considerably thinned, and only a few atrophic tubules remain (arrows) within an interstitium comprising dense fibrous tissue and a mononuclear cell infiltrate (blue staining nuclei). No glomeruli can be seen. This demonstrates why there would be no prospect for any significant functional recovery in this kidney even after the relief of the obstruction.

(patients on dialysis excluded), demonstrating significant but nonprogressive kidney impairment after relief of obstruction. It is likely that the prognosis for kidney functional recovery is better after earlier diagnosis and relief of the obstruction.

TREATMENT

General Considerations

Treatment is dictated by the location of the obstruction, the underlying cause, and the degree of any kidney impairment. If kidney impairment is present, the treatment of obstruction requires close collaboration between nephrologists and urologists to reduce the risks associated with the metabolic and electrolyte consequences of kidney failure and to optimize the chances for long-term recovery of kidney function. For example, complete bilateral ureteral obstruction (or obstruction of a single kidney) manifesting as AKI is a medical emergency and requires rapid intervention to salvage kidney function. Prompt intervention to relieve the obstruction should result in a rapid improvement in kidney function. Dialysis is rarely required in patients with AKI secondary to obstruction except to make the patient fit for intervention, such as by improving life-threatening hyperkalemia or severe fluid overload. The rapid relief of obstruction will limit permanent kidney damage, but kidney function may not recover immediately if acute tubular injury has resulted from obstruction or any accompanying sepsis.

Some surgical aspects of the management of obstructive uropathy are discussed in Chapter 63. The site of obstruction frequently determines the approach. If the obstruction is distal to the bladder, a urethral catheter or, if this cannot be passed, a suprapubic cystostomy will effectively decompress the kidneys. Placement of nephrostomy tubes or cystoscopy and passage of a retrograde ureteral stent will relieve upper urinary tract obstruction. Percutaneous nephrostomy (PCN) is generally the appropriate emergency treatment for upper urinary tract obstruction, especially in the setting of AKI. PCN can be achieved with local anesthetic and should allow for rapid recovery of kidney function

in most patients (>70%), avoiding the need for dialysis. After relief of the obstruction by a nephrostomy, the exact site and nature of the obstructing lesion can be determined by an antegrade study infusing radiographic contrast material into the nephrostomy tube (nephrostogram), and time can be taken to plan definitive therapy (Fig. 61.14). The benefit of PCN insertion is that it can be achieved under local anesthetic, whereas ureteral stenting usually requires a general anesthetic. However, PCN is usually not possible where patients are therapeutically anticoagulated, potentially limiting its use where it is required urgently, and anticoagulation cannot be reversed. Major complications of nephrostomy insertion (abscess, infection, and hematoma) occur in less than 5% of patients. If both kidneys are obstructed, the nephrostomy should initially be placed in the kidney with the most preserved kidney parenchyma, although bilateral nephrostomies may be required to maximize the potential for the recovery of kidney function. If infection occurs above a ureteral obstruction (pyonephrosis), drainage of the kidney by PCN can play an important therapeutic role together with appropriate antibiotics.

A nephrostomy can be used to gauge the potential for functional recovery in patients with chronic obstruction. Failure of kidney recovery after several weeks of nephrostomy drainage strongly suggests irreversible structural damage and thus no likely benefit from undertaking a more definitive surgical correction of the obstructing lesion. Long-term nephrostomy or stenting may be used as a definitive therapy for patients who are unsuitable for major surgical intervention and those with incurable malignant disease, although patient selection is important in the latter²³ (see Chapter 63 for further discussion). Various types of stent are available, and there is some evidence that metallic stents may provide more durable functional decompression in patients with malignant ureteric obstruction.²⁴

Where stents have failed or the ureters have become impassable because of disease, it is possible where expertise exists to insert “extra-anatomic stents.” These stents are tunneled from the kidney to the bladder percutaneously in the abdominal wall, thus bypassing any intraabdominal pathology.²⁵

Ureteral obstruction requiring intervention occurs in up to 3% of kidney transplant recipients.²⁶ It can be treated by nephrostomy and ureteral stenting, percutaneous incision or balloon dilation of the stricture, or open surgical repair (see Chapter 108).

Specific Therapies

Calculi are the most common cause of ureteral obstruction, and their treatment includes relief of pain, elimination of obstruction, and treatment of infection (see Chapters 60 and 63). Ureteral obstruction by papillary tissue, blood clots, or a fungus ball is treated by procedures similar to those used for calculi. When obstruction is caused by neoplastic, inflammatory, or neurologic disease, there is unlikely to be spontaneous remission of the obstruction, and some form of urinary diversion, such as an ileal conduit, long-term intermittent ureteral stenting, or extra-anatomic stenting should be considered. Some obstructing neoplastic lesions, such as lymphadenopathy from lymphoma, may respond to chemotherapy. A “realistic medicine” approach should be taken in management of ureteric obstruction because surgical management can reduce quality of life and carries risks of complications. Management of malignant urinary tract obstruction is discussed further in Chapter 59.

In idiopathic retroperitoneal fibrosis, ureterolysis (in which the ureters are surgically freed from their fibrous encasement) may be beneficial, especially if combined with corticosteroid therapy to prevent recurrence. The procedure can be performed minimally invasively using robotic technology in specialist centers and may represent a durable treatment preferable to long-term ureteric stents and steroids.²⁷

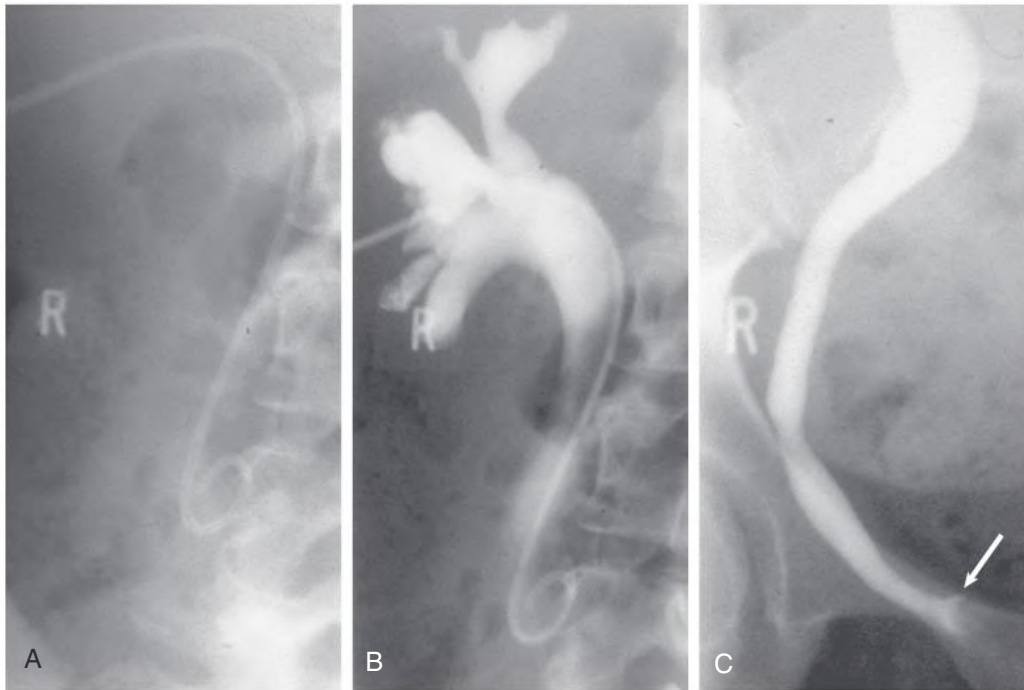


Fig. 61.14 Nephrostogram. A nephrostomy has been placed percutaneously into the dilated collecting system of the kidney under ultrasound control (A). After infusion of contrast material down the nephrostomy, the dilated pelvicalyceal system and upper ureter (B) and the lower ureter (C) are outlined. The ureter is dilated along its length but tapers abruptly at the vesicoureteral junction (arrow). In this case the obstruction was caused by a radiolucent stone.

Although ureteral stent insertion and corticosteroids can provide relief of obstruction in idiopathic retroperitoneal fibrosis,²⁸ these treatments come with long-term adverse effects, whereas surgery may offer the chance of cure.^{27,29} Retroperitoneal fibrosis complicating IgG4 disease may respond to corticosteroid treatment.³ Although kidney obstruction per se results in kidney inflammation involving multiple immune cells irrespective of the underlying cause, there are no published trials of therapies that target the immune system in this patient cohort. Although pharmacologic blockade of the renin-angiotensin system is not instituted in the acute setting, it is reasonable to employ ARBs or angiotensin-converting enzyme inhibitors in patients with residual CKD.

Functionally significant PUJ obstruction should be corrected surgically; minimally invasive and robotic techniques have largely replaced the open (Anderson-Hynes) pyeloplasty. The laparoscopic approach results in significantly less morbidity and has good long-term outcomes that are comparable to those of the open procedure.³⁰ Balloon dilation of the abnormal segment of the ureter is also possible, but the recurrence rate is high. The severity of kidney fibrosis and atrophy as assessed on intraoperative wedge kidney biopsy in patients undergoing open PUJ surgery can be used to predict functional outcome.³¹ There is increasing interest in the usefulness of urine proteomics to identify neonates with PUJ obstruction who require surgical intervention,³² although a subsequent small study indicated that this approach was less specific and sensitive in older children.³³ Urinary biomarkers that show altered levels in children or adult patients with obstructive nephropathy include established well-studied AKI biomarkers (kidney injury molecule-1, neutrophil gelatinase-associated lipocalin, and liver-type fatty acid binding protein),³⁴ aquaporin-2, and TGF- β , as well as epidermal growth factor, monocyte chemoattractant protein-1, and β_2 -microglobulin.³⁵ Interestingly, a reduction in urinary epidermal growth factor is also seen in patients with CKD and is a predictor of

CKD progression,³⁶ suggesting that urinary biomarkers may well facilitate the future stratification of patients with obstructive nephropathy and assist with predicting functional outcome after intervention.

Benign prostatic enlargement is the most common cause of lower urinary tract obstruction in males and may be mild and nonprogressive. Patients with minimal symptoms, no infection, and a normal upper urinary tract can continue with assessment until they and their physician agree that further treatment is desirable. Medical therapy with either α -adrenergic blockers (e.g., tamsulosin) or 5 α -reductase inhibitors (e.g., finasteride) may be used in patients with moderate symptoms.³⁷ α -Blockers relax the smooth muscle of the bladder neck and prostate and decrease urethral pressure and outflow obstruction. 5 α -Reductase inhibitors inhibit the conversion of testosterone to the active metabolite dihydrotestosterone and reduce prostatic hypertrophy. Combination therapy with these agents may be synergistic. Medical therapy is not appropriate for patients with outflow obstruction caused by BPE who have had upper urinary tract obstruction and loss of GFR. In these patients, definitive bladder drainage is mandatory and therefore will entail either long-term urethral catheter, intermittent self-catheterization, or definitive bladder outflow surgery according to individual patient risks and benefits. Surgical intervention for benign prostatic obstruction is generally required for failed medical treatment, debilitating symptoms, urinary retention, recurrent infection, or evidence of kidney parenchymal damage. A multitude of surgical modalities are available to treat benign prostatic obstruction, with transurethral resection of the prostate (TURP) still considered the gold standard; however, discussion with a urologist is recommended for all patients with significant BPE, even those who may traditionally have been considered “unfit for surgery.”³⁸

Urethral strictures in males can be treated by dilation or direct-vision internal urethrotomy before definitive urethroplasty, although urethral stenting is evolving. The incidence of bladder neck and

urethral obstruction in females is low and treatment rarely required. Suprapubic cystostomy may be necessary for bladder drainage in patients unable to void after injury to the urethra or in those who have an impassable urethral stricture.

When obstruction results from neuropathic bladder dysfunction, urodynamic studies are essential to guide therapy. The goals of therapy are to establish the bladder as a urine storage organ without causing kidney parenchymal injury and provide a mechanism for bladder emptying acceptable to the patient. Patients may have either a flaccid atonic or an unstable hypertonic bladder. Ureteral reflux and parenchymal damage may develop in both cases, although it is more common in patients with a hypertonic bladder. Asking the patient to void at regular intervals may achieve satisfactory emptying of the bladder. Patients with an atonic bladder and significant residual urine retention associated with recurrent urosepsis need to undertake clean intermittent self-catheterization. The aim should be to catheterize four or five times per day to ensure the amount of urine drained from the bladder on each occasion is less than 400 mL. In patients with a hypertonic bladder, improvement in the storage function of the bladder may be obtained with anticholinergic agents or intermittent detrusor injection with botulinum toxin. Occasionally, chronic clean intermittent self-catheterization is necessary.

Whenever possible, chronic indwelling catheters should be avoided in patients with a neurogenic bladder because they may lead to the formation of bladder stones, urosepsis, and urethral erosion, and they predispose to squamous cell carcinoma of the bladder. Patients who have chronic indwelling catheters for more than 5 years should have annual cystoscopic examinations. If deterioration in GFR occurs despite conservative measures or there is intractable incontinence or a small contracted bladder, an upper urinary tract diversion procedure such as an ileal conduit may be required or a bladder augmentation procedure such as a clam ileocystoplasty.

Management of Postobstructive Diuresis

Marked polyuria (postobstructive diuresis) is frequently seen after the release of bilateral obstruction or obstruction of a single functioning kidney. Indeed, a diuresis of greater than 7 L/day is associated with a good functional outcome.³⁹ Release of unilateral obstruction rarely results in a postobstructive diuresis despite the presence of tubular dysfunction and a concentrating defect. This is because of intrinsic differences in the tubular response to unilateral and bilateral obstruction and, more importantly, the salt and water retention and kidney

impairment that occurred in bilateral obstruction (not evident in unilateral obstruction because of the contralateral normal kidney). The resultant increase in natriuretic factors (including atrial natriuretic peptide) and substances able to promote an osmotic diuresis, such as urea, promote an appropriate postobstructive diuresis to excrete water and electrolytes that were retained during the period of obstruction. However, the postobstructive diuresis also may be inappropriate because of tubular dysfunction and, if not managed correctly, may result in severe volume depletion and electrolyte imbalance with continued kidney dysfunction. The key to correct management is clinical fluid assessment because appropriate diuresis may not require treatment, whereas inappropriate diuresis requires treatment to prevent hypovolemia and electrolyte imbalance. IV and oral fluid replacement is usually required in cases of inappropriate diuresis with careful and regular assessment of the fluid balance and serum electrolytes to tailor the fluid replacement regimen appropriately. Once the patient is deemed euvolemic, urine losses plus an allowance for insensible losses should be replaced. Urine volume should be measured regularly (hourly) to facilitate fluid administration, and serum electrolytes should be measured at least daily and as frequently as every 6 hours when there is a massive diuresis. Daily weighing of the patient is also helpful. Replacement fluid regimens should include sodium chloride and a source of bicarbonate and potassium. Calcium, phosphate, and magnesium replacement also may be necessary.

If fluid administration is overzealous, the kidney will not recover its concentrating ability, and a continued “driven” diuresis will result. It may then be necessary to decrease fluid replacement to levels below those of the urine output and observe the patient carefully for signs of volume depletion.

Future Prospects

Understanding the pathophysiologic changes that follow ureteral obstruction has allowed the development of rational interventional therapies to hasten the recovery of kidney function and limit permanent kidney damage. Although the best treatment option in humans remains prompt and effective relief of the obstruction, development and implementation of improved imaging modalities that provide more sophisticated anatomic and functional information (including intrarenal oxygen content⁴⁰) and future advances in urine proteomics will undoubtedly refine patient management and increase the data available for making key clinical decisions, such as whether and when surgical intervention is required.

SELF-ASSESSMENT QUESTIONS

- Which of the following statements are correct?
 - The presence of hydronephrosis does not always indicate significant obstruction.
 - Despite the effective relief of urinary tract obstruction, the majority of patients slowly progress to ESKD.
 - Urinary tract obstruction is always associated with oliguria.
 - Patients with urinary tract obstruction typically exhibit microscopic hematuria on urinalysis.
 - Urinary tract obstruction is a risk factor for urinary infections.
- Which of the following statements are correct?
 - The development of marked diuresis after relief of urinary tract obstruction is a poor prognostic indicator and is associated with the eventual development of ESKD.
 - Urinary tract obstruction may complicate neurologic disease.
 - Retroperitoneal disease may result in urinary tract obstruction.
 - Urinary tract obstruction leads to inflammation and scarring of the kidney.
 - Urinary tract obstruction may complicate IgA nephropathy.
- Which of the following statements are correct?
 - Defective tubular function may persist after relief of urinary tract obstruction.
 - Urinary tract obstruction is associated with a reduction in the glomerular filtration rate but preservation of kidney blood flow.
 - Non-contrast-enhanced spiral CT scanning is now established as the first-line imaging investigation in patients with suspected urinary tract obstruction.
 - A diuresis renogram accurately assesses kidney function but is unlikely to indicate the presence of obstruction.
 - Urinary tract obstruction may occur in the absence of hydronephrosis.

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